

CSMMECH000000004 Robot Redesign for Space Exploration

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | Date: June 2026 Category: Robotics / Space

Revision Drivers (all V2.0 documents)

- 1. 2025-2026 experimental and regulatory data supersedes V1.0 specifications
- 2. Solar Cycle 25 SSN ~137 peak (observed) — elevated urgency for all Carrington-focused work
- 3. Standards updates — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
Robot reference platform	Generic humanoid concept	SpaceX Optimus Gen 2 (2024): 57 kg specs	V1.0 lacked specific robot to design for
SPE radiation dose limit	Not quantified	Proton fluence 10^10/cm2 -> 50 kGy in copper windings	V1.0 missed single-storm failure mechanism
Mars dust GIC model	Conceptual	Triboelectric field >100 kV/m; ESD to frame 50 kJ	V1.0 did not model Mars-specific discharge mechanism
BFRP outgassing	ASTM E595 mentioned	Elum(r) 2025 space-grade qualification: TML<1.0%, CVCM<0.10%	V1.0 materials not certified for space vacuum
Actuator in vacuum	Electric motor vacuum issues noted	KNbO3-BaTiO3 DLC-coated stator: no air required	Electric motors have vacuum arc-flash risk; USM does not

Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.

End Versioning Delta | CSMMECH000000004 Robot Redesign for Space Exploration | June 2026